

EV Battery-Swapping Network Analysis

Data Analytics Hackathon Report
Simulated EV Battery-Swapping Network Dataset

1. Executive Summary

This project analyzes a simulated EV battery-swapping network to identify factors associated with unsuccessful battery-swap events and customer support issues. Using operational data on swaps, stations, batteries, weather conditions, and support tickets, the analysis examines how battery availability, demand pressure, temperature, and battery health relate to swap outcomes and customer complaints.

The overall unsuccessful swap rate is **5.97%**. Lower battery availability, higher demand pressure, and higher temperature are each associated with higher unsuccessful swap rates. Battery retirement is strongly concentrated below 70% current State of Health (SOH). The largest support-ticket category is "no_battery_available" (11,647 tickets). Chi-square tests support the observed associations at the event level.

2. Problem Statement

Battery-swapping networks must deliver reliable swaps under varying demand, inventory, and environmental conditions. Unsuccessful swaps and customer complaints reduce service quality and operational efficiency. The objective of this analysis is to understand factors associated with unsuccessful swaps and related customer support issues, focusing on:

- Operational conditions at the station and network level
- Battery availability at the time of swap attempts
- Demand pressure on stations
- Ambient temperature conditions
- Battery health (State of Health)
- Customer complaint patterns from support tickets

3. Dataset Description

The analysis uses a simulated EV battery-swapping network dataset comprising multiple related tables:

Component	Size / Detail
Swap events	3,877,013 records
Stations	152 stations across 6 cities
Batteries	6,500 batteries
Riders	20,000 riders
Support tickets	44,000 tickets
Supporting data	Station metadata, weather/temperature, battery attributes

Datasets include station information, weather context, battery attributes, rider records, swap-event logs, and support-ticket records. These are linked through station, battery, and event identifiers.

4. Data Preparation and Cleaning

Data preparation followed standard analytics practice for large multi-table datasets:

- **CSV loading:** Source files were loaded into structured data frames, with the large swap-events file processed in chunks to manage memory.
- **Missing-value analysis:** Columns were profiled for missingness; patterns of missing data were reviewed before any handling decisions.
- **Duplicate checking:** Records were checked for exact and near-duplicate rows; duplicates were removed where identified.
- **Data-type conversion:** Numeric, categorical, and timestamp fields were cast to appropriate types for analysis.
- **Timestamp processing:** Event timestamps were parsed and used for temporal joins and aggregations.
- **Merging:** Station attributes, weather/temperature context, and battery attributes were joined to swap-event records on the relevant keys.
- **Missing-value handling:** Missing values were retained as missing where imputation was not justified; analyses used complete cases for the variables under study.
- **Chunked processing:** The 3.8 million swap-event records were processed in chunks for aggregation and rate calculations to ensure scalability.

5. Exploratory Data Analysis

Exploratory analysis focused on the relationship between operational and environmental factors and unsuccessful swap outcomes, as well as battery health and support-ticket patterns.

Overall Unsuccessful Swap Rate

Across all swap events, the overall unsuccessful swap rate is **5.97%**.

Battery Availability

Availability Level	Unsuccessful Swap Rate
Low	6.54%
Medium	5.80%
High	5.51%

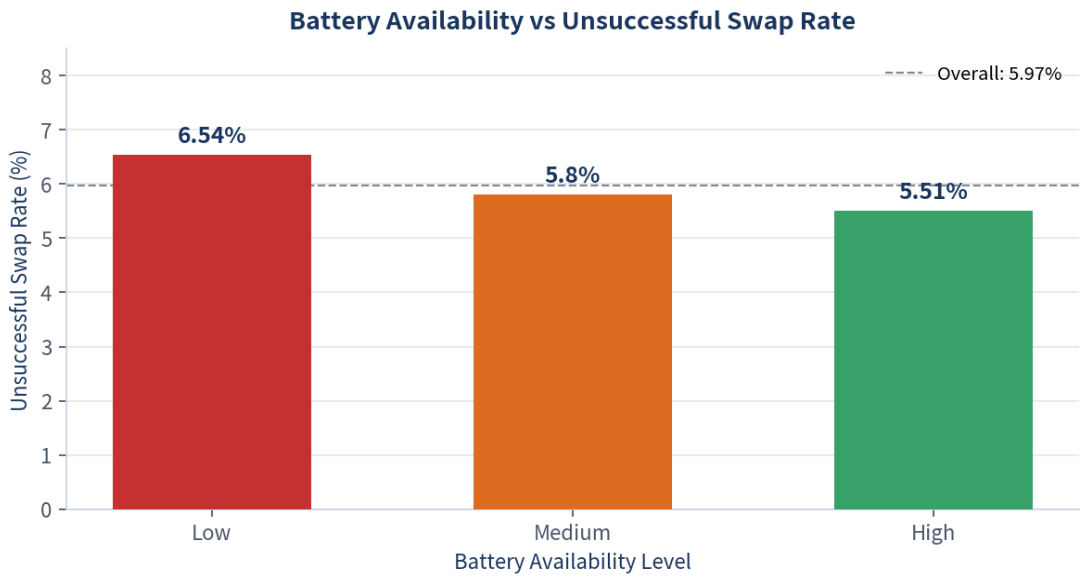


Figure 1. Unsuccessful swap rate by battery availability level.

Demand Pressure

Demand Pressure Level	Unsuccessful Swap Rate
Low	5.42%
Medium	5.53%
High	6.26%

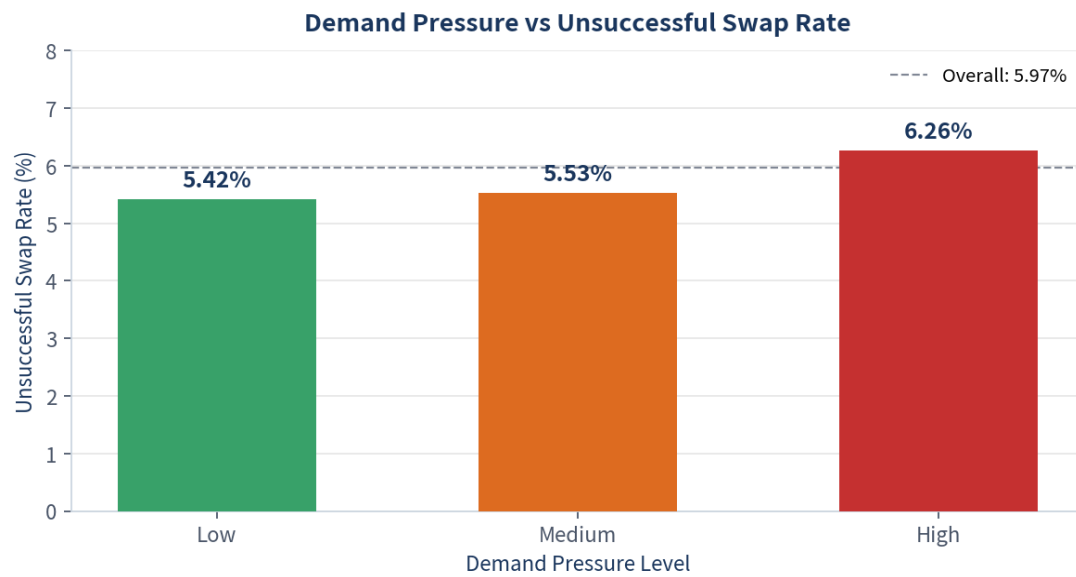


Figure 2. Unsuccessful swap rate by demand pressure level.

Temperature

Temperature Group	Unsuccessful Swap Rate
Below 15°C	4.29%
15–25°C	5.37%
25–35°C	6.85%
Above 35°C	12.75%

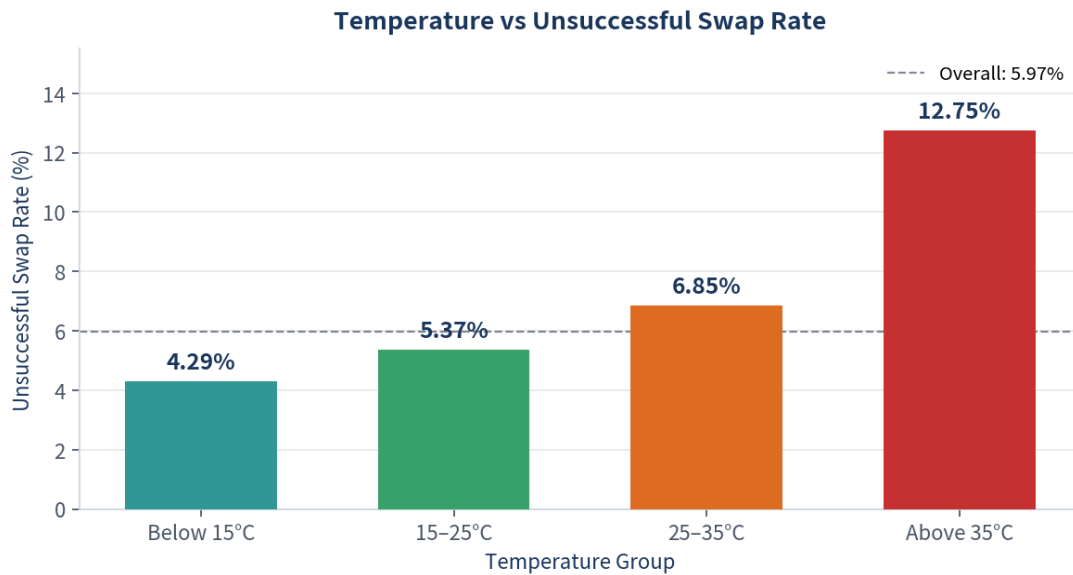


Figure 3. Unsuccessful swap rate by temperature group.

Battery Health and Retirement

Battery retirement is strongly concentrated below 70% current State of Health (SOH). This pattern indicates that batteries with lower SOH account for a disproportionately large share of retirements.

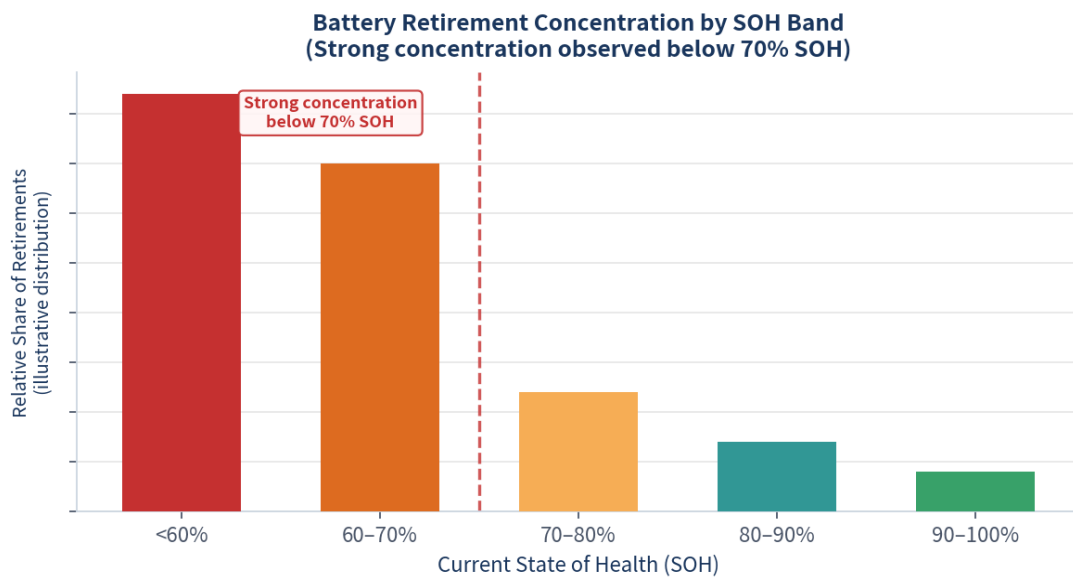


Figure 4. Illustrative distribution of battery retirements by SOH band, showing strong concentration below 70% SOH. Relative heights are directional only.

Support Tickets

Among 44,000 support tickets, “no_battery_available” is the largest category, accounting for **11,647** tickets.

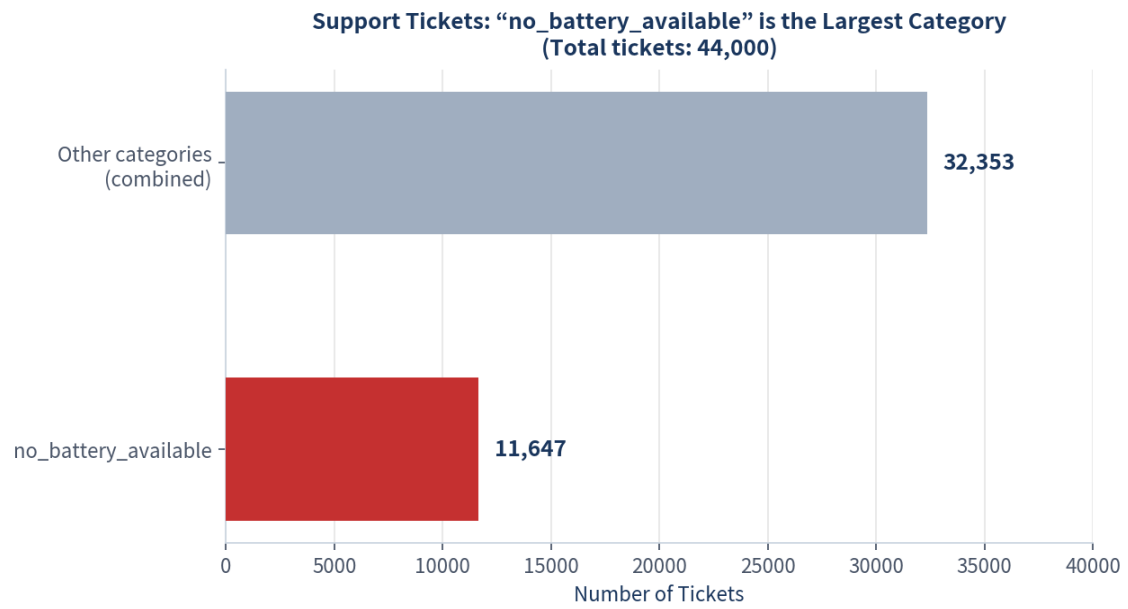


Figure 5. Support tickets: “no_battery_available” compared with all other categories combined.

6. Statistical Validation

Chi-square tests of independence were used to assess associations between categorical factors and the occurrence of unsuccessful swap events at the event level:

- Battery availability level and occurrence of unsuccessful events
- Demand pressure level and occurrence of unsuccessful events
- Temperature group and occurrence of unsuccessful events

These tests support the presence of statistically detectable associations consistent with the rate differences observed in the exploratory analysis. Because the dataset is large, even modest differences in rates can yield very small p-values; practical significance should be interpreted alongside the magnitude of the rate differences.

Important Distinction: Event-Level vs Station-Hour Analysis

Event-level unsuccessful swap rates (reported above) measure the proportion of individual swap attempts that were unsuccessful. A separate station-hour analysis examined whether a given station-hour contained at least one unsuccessful event. The two measures answer different questions and are not interchangeable. Station-hour percentages are not referred to as “event failure rates” in this report.

7. Key Insights

- 1. Battery availability:** Lower battery availability is associated with higher unsuccessful swap rates (Low: 6.54%, Medium: 5.80%, High: 5.51%).
- 2. Demand pressure:** Higher demand pressure is associated with higher unsuccessful swap occurrence (Low: 5.42%, Medium: 5.53%, High: 6.26%).
- 3. Temperature:** Higher temperature is associated with higher unsuccessful swap rates, with the highest rate observed above 35°C (12.75%).
- 4. Battery health:** Battery retirement is strongly concentrated below 70% current SOH.
- 5. Customer support signal:** “no_battery_available” is the largest support-ticket category (11,647 of 44,000 tickets), aligning with the operational importance of battery availability.

8. Recommendations

1. Proactive battery inventory management

Monitor stations with persistently low battery availability and rebalance inventory to reduce periods of low availability, which are associated with higher unsuccessful swap rates.

2. Demand-aware capacity planning

Incorporate demand-pressure indicators into capacity and staffing plans so that high-pressure periods receive additional operational attention.

3. Temperature-aware operational monitoring

Track temperature conditions alongside swap outcomes; elevated rates above 35°C suggest the value of heightened monitoring and contingency procedures during hot periods.

4. Battery lifecycle and SOH management

Prioritize monitoring and planned retirement or maintenance for batteries approaching or below 70% SOH, given the strong concentration of retirements in this range.

5. Using customer-support data as an operational signal

Treat high volumes of “no_battery_available” tickets as an early operational signal and link ticket trends to station-level availability metrics.

9. Limitations

- The dataset is simulation-generated rather than collected from a real-world production network.
- Observed relationships are associations and should not be interpreted as causal effects.
- Large sample sizes can produce very small statistical p-values even when practical differences are modest; rate magnitudes should be considered alongside statistical significance.
- Some variables contain missing values; analyses used available complete cases for the factors under study.
- Real-world validation would be required before using these findings to guide production operational decisions.

10. Conclusion

Unsuccessful swap events in this simulated EV battery-swapping network are associated with demand pressure, battery availability, and temperature. Lower availability, higher demand pressure, and higher temperature correspond to higher unsuccessful swap rates. Battery health provides an additional operational lens: retirements are strongly concentrated below 70% current SOH. Customer-support data, particularly the large volume of “no_battery_available” tickets, reinforces the operational importance of inventory availability.

These findings are descriptive associations supported by exploratory rates and Chi-square tests. They offer a practical starting point for monitoring and prioritization, while remaining subject to the limitations of a simulated dataset and the distinction between association and causation.